

GonitSathi Paper Preparation Guideline

Research plan for student research assistants

Prepared for the GonitSathi student RA team | 12 September 2026

Target venue: OE-Agent at ACML 2026

This guideline turns the GonitSathi-BCS idea into a focused, testable paper about reliable tutoring with small open-weight models. The recommended contribution is an evidence-governed diagnostic controller: it records what the student actually did, keeps competing explanations for an error, controls when a learner-state claim can become active, and spends a limited interaction budget on a diagnostic question when that question is useful.

The central experiment must show whether this controller improves diagnostic reliability at a comparable level of diagnostic coverage and computational cost. A fluent Bengali tutor, a symbolic solver, a learner graph, and multiple agents are useful engineering components, but their presence alone does not establish this contribution.

Use this document as an execution manual. Read Sections 1-5 together before implementation; assign ownership using Section 15; freeze the evaluation protocol before examining held-out results. Every proposed sample size, parameter setting, and acceptance criterion below is a planning recommendation, not a completed experiment or an established property of GonitSathi.

The submission should distinguish three kinds of evidence: results reported for the earlier engine, results newly reproduced for the submitted system, and experiments still planned. The existing report is a design foundation; it is not yet evidence that the proposed tutor improves student learning.

The paper in one sentence

We investigate whether provenance-aware diagnostic state updates and bounded information-seeking interactions make small open-weight tutoring agents more reliable and efficient on ambiguous Bengali mathematics attempts.

What the team must deliver

1. A narrow contribution statement and a primary-source comparison with the closest prior work.
2. A reviewed benchmark with family-disjoint splits, ambiguous cases, intervention provenance, and hidden diagnostic labels.
3. A reproducible controller, strong matched baselines, and a complete experiment registry.
4. Diagnostic risk and coverage results, calibration, useful intervention quality, and measured resource costs.
5. Mechanism ablations that distinguish evidence governance from extra calls, extra student responses, conservative abstention, and verified answer access.
6. An anonymous manuscript, reproducibility appendix, and a release package that another RA can run independently.

Reading map

Sections	Purpose
1-2	Workshop requirements and corrections to the source report
3-4	Recent prior work, novelty boundary, hypotheses, and scope
5-6	Implementable method and a worked diagnostic example
7-8	Dataset construction, annotation, and baseline definitions
9-12	Run settings, metrics, experiments, and ablations
13-14	Human evaluation, statistics, and interpretation
15-17	RA schedule, manuscript plan, and submission gates
Appendices	Event schema, prompt contracts, result templates, and references

1 Workshop requirements and submission strategy

1 1 Verified requirements

The workshop focuses on open and efficient agentic AI under limited compute and deployment resources. Its relevance to GonitSathi comes from using verified tools, compact state, and selective computation to support capable tutoring with small models. The following details were checked against the workshop website on 12 September 2026. [W1]

Item	Requirement or current listing
Full paper	Up to 8 pages, excluding references and appendices
Short paper	Up to 4 pages; preliminary results and emerging ideas are welcome
Review submission	Anonymous; official ACML 2026 LaTeX template
Submission deadline	18 October 2026, 23:59 Anywhere on Earth
Bangladesh equivalent	19 October 2026, 17:59 Bangladesh time
Internal target	Submit by 16 October 2026 Bangladesh time
Notification	30 October 2026
Camera ready	13 November 2026
Workshop website date	1 December 2026; exact session times pending
Review criteria	Relevance, originality, technical quality, clarity, reproducibility, impact

Use the workshop's linked OpenReview group: [OE-Agent submission portal](#). Use its linked [ACML template archive](#), but configure the review PDF for anonymity. The archive name does not override the workshop's anonymous-submission requirement. [W1]

Before final upload, the submission lead must recheck the portal's abstract deadline, supplement limits, reference treatment for short papers, and anonymity fields. The workshop page does not explicitly settle archival publication or concurrent-submission

policy. Do not infer these from the main conference. The portal listing and workshop website have displayed different event dates; use the website's 1 December listing provisionally and verify attendance scheduling later. These administrative uncertainties do not prevent preparing the paper. [W1, W2]

1 2 Recommended paper scope

Aim for an eight-page empirical methods paper. Lead with small-model diagnostic reliability and show an efficiency trade-off. The mobile app, subscriptions, accounting/calculus engines, unrestricted RAG, and a general educational platform should not consume the main contribution or development schedule.

A full paper is appropriate if the team completes a reviewed test set, a reproducible open-model implementation, credible close baselines, and the primary mechanism tests. If only a smaller benchmark and preliminary experiment are ready, submit a four-page paper with narrower claims. A short paper is an acceptable scientific outcome; unsupported claims of long-term learning are not a substitute for missing evidence.

The roughly five-week window before submission cannot accommodate a newly started six-to-eight-week intervention followed by four- and eight-week retention assessments. Keep the longer study as a subsequent project unless those data already exist and were collected under an appropriate protocol.

2 Audit the source report before implementation

2 1 Preserve useful assets and separate their evidence

The uploaded GonitSathi Research Report v2 describes an earlier semantic compiler, symbolic tools, a cache, step verification, and the proposed tutoring workspace. It reports 12% GSM8K accuracy on 50 items, 90% verifier accuracy, exact-cache latency of 1.25 ms, and perfect calculation precision within tested supported cases. These are source-reported historical results. Recover their datasets, split definitions, scripts, hardware, and raw logs before including them as reproducible evidence. [S1, pp. 2-3]

A correct solver can execute the wrong equation perfectly. The submitted system therefore needs separate measures of problem interpretation, step alignment, symbolic execution, and released explanation correctness. Likewise, a cache result on exact inputs does not establish general tutoring efficiency.

For each inherited number, create a claim ledger with the original wording, denominator, task, dataset version, system commit, supporting file, and decision to reproduce, qualify, or remove. If a number cannot be reconstructed, use the earlier engine only as development motivation.

2 2 Required methodological corrections

Source report issue	Required correction
Verification factor becomes zero when the student's arithmetic is wrong	Distinguish an untrustworthy observation from a reliably verified error. Wrong work can strongly support an error hypothesis.
Product of confidence, independence, source, difficulty, and recency is treated as diagnostic reliability	Treat this as an uncalibrated heuristic baseline until validated against held-out evidence. Factors need operational definitions.

Source report issue	Required correction
Taking logarithms is presented as fixing product decay	Log-space arithmetic helps numerical computation. It does not change the product's statistical meaning or calibrate it. Zero factors require separate handling.
An entropy-based probe selector lacks a response model	Specify how response probabilities are estimated, how missing outcomes are handled, and how the selected probe is evaluated.
Confirmed learner traits appear permanent	Make claims context-specific, revisable, and time-stamped; keep retraction and contradiction records.
One canonical solution graph is treated as mathematical truth	Accept valid alternative strategies and equivalent expressions. Distinguish unsupported input from an incorrect step.
Mathematical checks are treated as a guarantee for generated prose	Recheck the released explanation, including units, assumptions, and disclosure policy. Report residual errors.
Broad literature claims include precise correlations and percentage reductions	Retain only claims tied to a verified table, metric, and experimental setting in the original paper.

Do not repeat the report's claims that researchers universally agree, that symbolic delegation eliminates every arithmetic failure, that one scalar necessarily distorts all student models, or that tokenization is the established cause of every Bengali performance gap. These require evidence beyond the supplied synthesis. BKT is a meaningful baseline; it is not inherently invalid because it represents a mastery probability.

2 3 Define the units that the system models

An observation is a recorded student action, such as an equation, a selected option, or a response to a probe. A verification result says whether the interpretation and mathematical relation are supported. A diagnostic hypothesis proposes an explanation, such as a percentage-base misconception or a transient slip. A learner-state claim is a contextual assertion supported by accumulated evidence. An instructional action is a choice of clarification, probe, hint, feedback, practice, or a requested worked example.

These objects must be distinct in the implementation. A verified incorrect equation does not prove a persistent misconception. A correct answer following a strong hint does not prove independent mastery. An LLM-generated diagnostic label is a candidate inference, not a direct measurement of the student's mind.

3 Recent related work and the novelty boundary

3 1 Required comparison matrix

Read the methods and evaluation sections of the closest papers, not only their abstracts. The comparison below identifies relevant prior art and the experiment GoniT Sathi needs. An absence from a paper's stated evaluation is not proof that its architecture cannot support that feature.

Prior work	Already established	GonitSathi comparison required
IntelliCode, EACL 2026 [R1]	Versioned learner state, single-writer validation, BKT-inspired updates sensitive to hints and other context	Compare against a strong validated-update baseline, not only unrestricted LLM writes
ScaffoldLM, ACL 2026 [R2]	Step plans and an assessment-driven memory loop	Test whether explicit evidence admission and diagnostic abstention add value to plan-based tutoring
SLOW, AIED 2026 [R3]	Explicit evidence parsing and cognitive validation in a workspace separating diagnosis from action	Compare observed diagnostic probes with internal counterfactual reasoning at matched resources
Auditable Release Control, August 2026 preprint [R4]	Separate disclosure decisions and an auditable release boundary	Treat output control as supporting infrastructure; measure usefulness as well as leakage
Stepwise Verification, EMNLP 2024 [R5]	First-error annotation and verification-grounded feedback	Hold first-error information constant when testing state governance
StratL, ACL Findings 2025 [R6]	Prompt optimization for transition-graph pedagogical plans	Include a fixed pedagogical controller with the same content and model
Pedagogical RL, EMNLP 2025 [R7]	Tutor specialization through simulated interaction and RL	Separate controller gains from model training; use a released tutor only if reproducible
MathTutorBench, EMNLP 2025 [R8]	Broader tutoring evaluation beyond solving accuracy	Use pedagogical quality and longer dialogues as evaluation dimensions

3 2 The three closest architectural comparisons

IntelliCode is particularly important because a single authorized state writer is already prior art. Its controller validates proposed updates, and its learner model already adjusts for assistance. Our proposed distinction is the measurement of evidence sufficiency for a particular diagnostic claim, including conflicting observations, selective commitment, and a bounded opportunity to collect new evidence. Present this as an empirical extension to test, not a claim that prior systems lack governance. Inspect its Sections 2.1 and 3 when implementing the comparison. [R1]

ScaffoldLM maintains step plans, progress, inferred learner states, and interaction history through an assess-act-track-record loop. Our candidate contribution concerns the reliability of the state estimate before it drives the next action. A plan-memory implementation plus the proposed evidence controller is therefore a useful incremental comparison. If we reproduce only the inference architecture, we must label it as an adaptation rather than as the original trained system. [R2]

SLOW already makes diagnostic reasoning explicit and uses counterfactual cognitive validation. The relevant distinction is whether the tutor obtains a real additional response when competing explanations remain unresolved. Internal simulation is not a student observation. Compare the two ways of reducing uncertainty while accounting for their different computation and interaction costs. SLOW is listed in the AIED 2026 proceedings; the report's description of it only as a concurrent preprint should be updated. [R3]

3 3 Bengali reasoning and efficiency prior art

Work	Relevance	Boundary for our paper
BanglaMATH, MathNLP 2025 [R9]	Bengali school mathematics with robustness and language evaluation	Bengali math and distractor evaluation are established
BenNumEval, ACL Findings 2025 [R10]	Multiple Bengali numerical reasoning tasks	Use applicable tasks for external solving checks; it does not supply learner-state labels
GanitLLM, 2026 [R11]	Bengali model and curriculum-based training; arXiv v3 reports ACL Findings acceptance	Include a specialized model where available; Bengali adaptation alone is insufficient novelty
DAGGER, January 2026 [R12]	Executable graph reasoning with distractor handling and token-efficiency evaluation	Graphs, Bangla distractors, and reduced tokens are already a close combination
GSM-Plus-BN, July 2026 preprint [R13]	Bengali perturbation benchmark	Do not claim the first Bengali perturbation benchmark
Small Language Models are the Future of Agentic AI, 2025 [R14]	Position paper motivating small-model agent systems	Motivation is not proof of our tutor's performance
RouteLLM, ICLR 2025 [R15]	Routing between models for cost and quality	Routing alone is not our algorithmic contribution
PAL, ICML 2023 [R16]	Executable programs separating language interpretation and computation	Symbolic tool use is inherited methodology

GonitSathi's benchmark can add something distinct if it annotates student attempts, competing error explanations, assistance provenance, evidence sufficiency, and useful diagnostic responses. A question-answer collection translated into Bangla does not establish that contribution.

3 4 What can support a contribution

The recommended contribution has three connected parts. First, implement an observation ledger and diagnostic controller that prevents unsupported state commitments while preserving evidence of both errors and successes. Second, evaluate a bounded probe policy that chooses whether additional evidence is worth the student's time. Third, release an evaluation protocol that tests these decisions under matched compute, interaction, and answer-access conditions.

The publishable finding might be that assistance-aware evidence improves selective diagnosis even with a small model. It might instead be that a simple Bayesian controller matches a complex agent, or that fixed templates dominate expensive generation for the supported task. These are useful findings if the experiments are rigorous and the manuscript reflects them.

Do not claim the invention of hypothesis-based diagnosis, Bayesian updating, information gain, contradiction-aware student modeling, symbolic verification, or learner memory. The earlier report itself points to classical intelligent tutoring work on diagnostic hypotheses and contradictory knowledge. [S1, pp. 10-11; R17]

3 5 Literature review assignment

For each close paper, one RA must record: exact version and venue; problem definition; accessible code and license; state variables; update rule; training requirements; supervision; access to reference solutions; evaluation unit; human versus simulated evidence; cost reporting; and the smallest defensible difference from GonitSathi. A second RA must verify the difference against the method section.

Recheck the literature on 5 October and before the final manuscript freeze. Use title searches, citation links from the closest papers, ACL Anthology, arXiv, and author repositories. Include new papers that materially overlap with our mechanism even if they weaken the initial novelty story. Keep preprints labeled and do not copy their comparative claims without checking the underlying experiment.

4 Research questions and a controlled scope

4 1 Main question and hypotheses

ID	Testable question	Primary evidence
RQ1	Does evidence governance reduce unsupported diagnostic commitments at comparable coverage?	Risk-coverage curve; risk at a frozen operating point; calibration
RQ2	Does a selected probe outperform an equally costly generic or random probe?	Diagnostic change after an observed response; inappropriate intervention rate
RQ3	Does the approach offer a useful quality-cost trade-off across small model sizes?	Accepted intervention quality against time, tokens, calls, and memory
RQ4	Does it resist assistance contamination, contradictory evidence, and language perturbations?	Paired stress tests and recovery after correction
RQ5	Does it help actual learners?	Blinded expert judgments and, separately, a properly designed learner study

Primary directional hypothesis: compared with the strongest matched state-memory baseline, the proposed controller lowers unsupported-commit risk while retaining useful diagnostic coverage. A method that never commits anything cannot satisfy this hypothesis.

Secondary hypotheses: assistance-conditioned updates reduce false mastery after hints; selected probes improve diagnosis more than equal-budget generic probes; a small model with the controller reaches a better measured efficiency trade-off than an always-generating pipeline. These are hypotheses to test, not predictions to insert into an abstract as findings.

4 2 Initial domain

Limit the first benchmark to six BCS-relevant topic groups: percentages and profit/loss; ratios and proportions; work and time; speed and distance; averages and mixtures; elementary algebra and number relations. Verify topic coverage against the materials used by the team and document excluded areas. Geometry diagrams, handwriting, speech, OCR, and open-ended proofs can be future extensions.

Support typed Bengali, Arabic and Bengali numerals, ordinary mathematical notation, and controlled mixed-language attempts. Begin with one problem at a time and at most six tutor-student exchanges per episode. A learner may request a worked solution or end the interaction. The policy must support these legitimate actions without labeling all answer disclosure as leakage.

4 3 Scope by priority

Priority	Required components
Essential	Reviewed problems; robust step checks; structured evidence; state controller; open model; matched baselines; risk/coverage; cost logs
Essential for probe claim	Reviewed candidate probes, a response model, actual observed probe outcomes, equal-interaction controls
Strong addition	Independent expert audit, multiple model sizes, 20-turn state stress tests, measured offline behavior
Later extension	Six-to-eight-week RCT, long retention follow-up, production mobile app, training a new tutor model

5 Implement the diagnostic controller

5 1 State and module boundaries

Use one controller with explicit tools. Separate responsibilities in code, but do not add LLM calls merely to create more named agents. The main pipeline is: record the student event; normalize and align the attempt; verify the relation; extract candidate evidence; update a diagnostic belief; decide whether to probe or teach; generate or select a permitted response; audit and release it.

Maintain four stores: an immutable observation log, a versioned problem and solution bank, a hypothesis ledger, and an active learner-state view. A graph can be implemented in relational tables. The graph data structure itself is not the novelty; its value must appear in traceability, selective diagnosis, or recovery measurements.

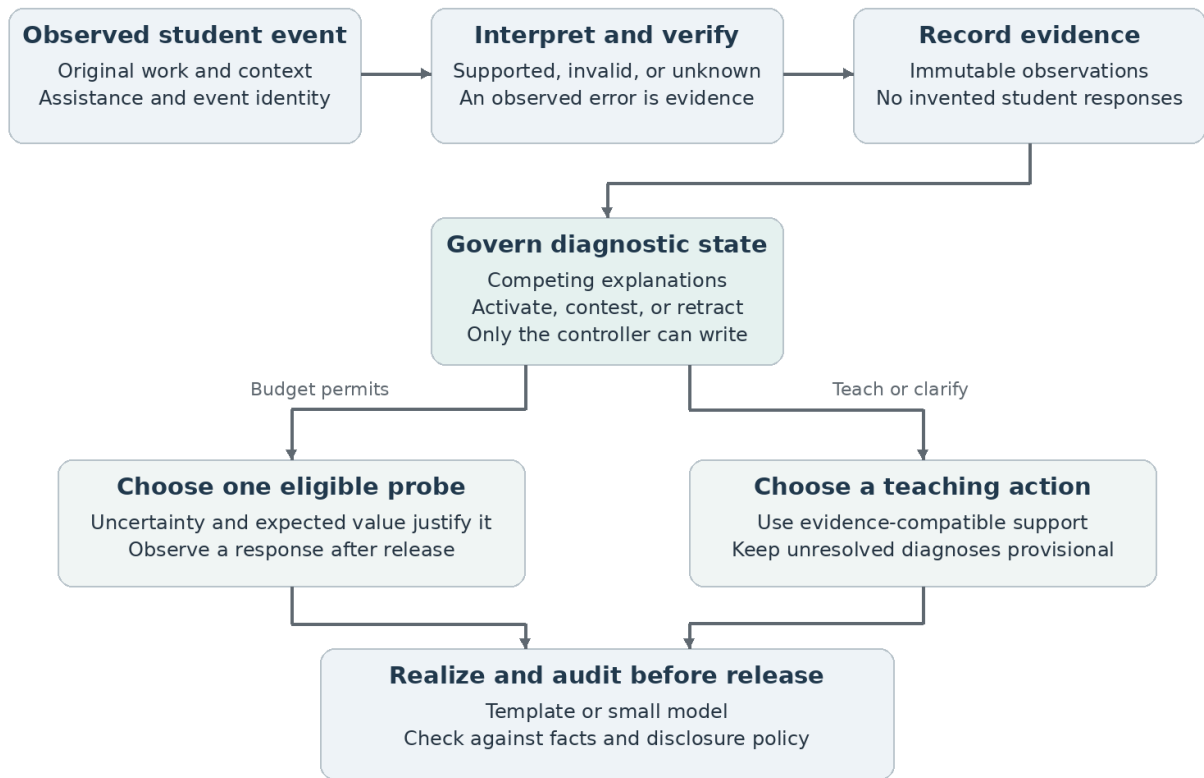


Figure 1. The controller separates observations, diagnostic claims, and instructional actions. A new student response starts another pass through the same evidence boundary.

Each hypothesis record needs a learner pseudonym, skill, error code, context, supporting event IDs, contradicting event IDs, assistance status, probability or score, creation time, latest update, model/controller version, and status. Use statuses such as provisional, active, contested, and retracted. Keep raw observations after retracting a claim.

5 2 Normalize and verify without erasing ambiguity

Keep the original text alongside normalized text. Convert numeral glyphs while preserving quantities, percentage signs, units, names, and the direction of comparisons. Store a reversible mapping between the two representations. Test ambiguous decimal notation and mixed numerals explicitly.

Parse only a restricted mathematical representation. Check variable binding, units, domain assumptions, and equivalence before using symbolic execution. Do not execute arbitrary generated Python as the student's trusted mathematical interpretation.

Return separate fields for interpretation confidence and mathematical status. Mathematical status should include valid, invalid, unverifiable, and unsupported. An unrecognized but potentially valid strategy is not automatically an error. For noncanonical strategies, try symbolic equivalence or request clarification. Report alternative-solution acceptance separately from the supported-template accuracy.

5 3 Evidence admission

Admit an observation to the diagnostic update only when its provenance and interpretation are adequate for that update. A trusted arithmetic error remains an admitted observation with status invalid. A corrupted transcript or unresolved interpretation may remain in the log without changing a diagnostic belief.

Treat assistance as part of the observation model. Independent correct work, a response after a conceptual hint, a response after the key equation was supplied, and a copied final

answer provide different evidence. Do not count them as equivalent mastery observations. Still allow assisted work to inform what support currently helps the learner.

Group correlated evidence. Several paraphrases of one attempt, repeated tool evaluations of the same equation, or duplicated sync events are not independent observations. Use an event identifier and a source-family identifier to prevent double counting. Where a model-derived feature comes from the same attempt as a correctness feature, model the joint response pattern instead of multiplying two independent-looking likelihoods.

5 4 A small probabilistic model before a complex scorer

For the workshop implementation, start with a categorical belief over plausible causes within one skill and episode. Candidate states can include a specific conceptual misconception, arithmetic slip, language misunderstanding, interface/transcription slip, no observed misconception, and unresolved/other. Restrict the candidate set using the current problem and observed trace.

For each candidate, combine its prior probability with the likelihood of the observed response pattern under that candidate, the problem, and the assistance condition; then normalize across candidates. Estimate likelihoods from the training partition with smoothing and expert-reviewed response categories. Use the development partition for calibration and policy thresholds. This is an operational diagnostic model, not a claim to recover a person's true internal mental state.

If observations within an episode are not conditionally independent, use a sequence-level feature model or count only the newest independent evidence group. For a short diagnostic episode, a static cause model is a manageable approximation. Across instruction and later sessions, use an explicit transition model or reopen the diagnosis; do not accumulate old likelihoods indefinitely as if learning cannot occur.

Use a small calibrated logistic or multinomial classifier as an alternative. This is often easier to validate than a six-factor product. Train only on training records, calibrate only on development records, and freeze coefficients before test evaluation. A verbal confidence score from the same LLM is a feature to validate, not an automatically calibrated probability.

The product rule from the source report can be retained as an ablation. Define every factor, its range, handling of zeros, and weight. A normalized weighted geometric mean may change its scale, but it still requires empirical validation. The manuscript must not describe numerical stability as statistical calibration.

5 5 Rules for activating and revising claims

Require sufficient posterior support, adequate observation quality, and sufficient independent evidence before activating a persistent misconception claim. A useful initial rule is at least two independent, relevant, unassisted evidence groups for a persistent claim. An episode-specific provisional diagnosis may be used earlier to select a gentle hint. Validate this rule against a one-observation baseline.

Sweep activation thresholds of 0.70, 0.80, 0.90, and 0.95 on development data. Choose a primary point using the risk and coverage rule in Section 10. Test a separate retention threshold below the activation threshold to prevent rapid oscillation. Thresholds are policies, not universal constants.

When a later observation contradicts an active claim, mark it contested and lower its operational authority. Retraction must cite the new evidence and retain the earlier record. Do not erase conflicting observations or treat a revision as a logging failure. Repeated independent success on a new problem is more informative than reciting an answer just supplied by the tutor.

5 6 Diagnostic probe selection

Use a small expert-reviewed candidate bank, initially two to six probes for each common ambiguity. A probe should discriminate among explanations without doing the target problem's decisive work for the student. For example, a direction-of-change question, a request to state the assumed relationship, or an independent microproblem may distinguish causes.

For each candidate probe, estimate possible response categories and their likelihood under each hypothesis. Expected information gain is the current diagnostic entropy minus the expected entropy after a response. The expectation must use probabilities for every response category, including incorrect, ambiguous, skipped, and timeout outcomes. Document smoothing and handling of rare categories.

Select the eligible probe with the largest expected information gain after subtracting normalized burdens for response time and inference cost. Reject probes that violate disclosure policy. Use a bounded lookup over the reviewed bank; avoid generating a large fresh candidate set for every turn unless that extra cost is part of the experiment.

Only probe when uncertainty is high, a candidate has sufficient expected value, and the interaction budget permits it. Normalize entropy by the base-two logarithm of the number of candidate classes; use zero if only one class remains. Sweep normalized thresholds of 0.4, 0.6, and 0.8 and maximum probe counts of 0, 1, and 2 on development data. The default is at most one diagnostic probe per problem; the two-probe condition is a sensitivity experiment.

After the student responds, update the belief with that actual response. Never update a persistent learner claim using a response predicted by the LLM or simulator inside the planner. If uncertainty remains, provide a neutral, local scaffold without asserting a persistent misconception. Record that the diagnosis remains unresolved.

5 7 Action selection and response release

Use a small action vocabulary: ask for clarification; ask a diagnostic probe; acknowledge a correct step; give a conceptual hint; offer a local scaffold; request an independent retry; offer a transfer question; provide an authorized worked solution; or defer an unsupported problem.

Give the language generator only the selected action, necessary verified facts, current student context, and allowed disclosure level. It may propose response text and cited fact IDs. It must not write learner state or authorize itself to disclose a solution.

Audit each final response for factual consistency and policy fit. In a protected hint state, leakage includes the final answer, the decisive calculation, or cumulative hints that effectively complete the solution. In an explicitly authorized worked-example state, those disclosures are allowed. A deterministic check can detect some violations; it cannot certify all semantic implications. Use an expert-reviewed fallback when uncertain and report its frequency and usefulness. [R4]

5 8 Runtime cost and agency

Use rules and cached reviewed hints when they are adequate. Invoke a small model for interpretation or realization only when needed. An optional larger-model route must be measured separately and must not silently rescue the nominally local-only condition.

Report the controller's choices and feedback loop to establish agentic behavior: it observes an attempt, selects a tool or interaction, receives new evidence, updates its belief, and changes its next action. A fixed sequence of prompt calls without state-dependent choices is a weaker basis for an agentic contribution.

The cost of evaluating probes grows with candidate probes, diagnostic hypotheses, and response categories. Account for all three, plus response-model or LLM costs. Do not report only a small entropy calculation while excluding the calls required to obtain its inputs.

6 Worked example for implementation and annotation

6 1 The observed error

Problem: the price of rice increases by 25%. By what percentage must consumption decrease to keep expenditure unchanged? If initial price and quantity are both normalized to 100, the new quantity is 80 and the decrease is 20%. This is the reference solution for the tutor and evaluator; it need not be exposed to the student immediately.

The student answers 25%. That answer alone supports several explanations: using the wrong percentage base, a transient arithmetic or transcription error, misunderstanding the direction of change, or guessing. Label the evidence as insufficient to identify a persistent misconception unless additional work clarifies the cause.

6 2 A permissible diagnostic interaction

Ask: If each kilogram becomes more expensive while the total budget stays fixed, should the number of kilograms bought increase, decrease, or stay the same? Briefly explain.

This question distinguishes some directional misunderstandings without revealing the target decrease. It does not adequately separate every percentage-base misconception from a slip. The response model and expected information gain must reflect that limitation. A second possible probe could use a different simple situation about preserving a product, provided experts judge that it does not give away the current answer.

If the response supports the correct direction but not the correct quantitative relationship, keep the percentage-base and slip hypotheses open. Offer the local hint: write the relationship among price, quantity, and total expenditure. Ask the student to continue. Do not store a permanent label merely because the original option was wrong.

6 3 Numerical information gain example

For an illustrative two-hypothesis case, suppose the prior is 0.5 conceptual error and 0.5 transient error. Suppose a reviewed probe has a 0.2 probability of a correct response under the conceptual-error hypothesis and 0.8 under the transient-error hypothesis. These probabilities are invented teaching values, not estimated GonitSathi parameters.

The current entropy is 1 bit. Correct and incorrect responses each have probability 0.5. Either response gives a posterior of 0.2/0.8 in one order, whose entropy is approximately 0.722 bits. Expected information gain is therefore approximately 0.278 bits. A less diagnostic probe with equal response likelihood under both hypotheses has zero expected information gain.

This calculation predicts uncertainty reduction under the assumed response model. It does not establish a learning benefit, and it can be wrong when the response model is wrong. Test sensitivity to perturbed likelihoods and compare predicted gain with observed diagnostic improvement.

6 4 The assistance trap

If the tutor supplies the equation and the student substitutes the numbers correctly, record a successful assisted step. Do not treat it as independent proof that the student knows how to formulate the equation. Later ask an equivalent but distinct problem without the

equation. That response can support an independent claim. This paired situation is a required benchmark case and ablation.

7 Construct GonitSathi Bench

7 1 Three evaluation layers

Build separate evaluation layers because a final-answer dataset cannot validate every part of a tutor. Layer A evaluates mathematical interpretation and first-error localization on fixed attempts. Layer B evaluates diagnostic belief and state updates over controlled histories with known evidence provenance. Layer C evaluates interactive action selection with actual responses to the selected probes. Human learning is a fourth, separate outcome requiring a learner study.

For Layer C, a static transcript cannot reveal what a student would have answered to an unasked question. Use either a reviewed branching response bank with hidden scenario state, or collect real interactions. An LLM simulator may add variety, but its outputs remain simulation evidence.

7 2 Core and expanded dataset budgets

Dataset component	Core workshop target	Expanded target
Independent problem families	300 across six topic groups	600 across the same groups
Train / development / test family targets	150 / 75 / 75	300 / 150 / 150
Fixed attempt histories per family	4	4
Total fixed histories	1,200	2,400
Held-out fixed histories	300	600
Interactive scenarios from test families	120 across at least 60 families	240 across at least 120 families
Language stress subset	100 held-out histories with 3 paired variants	200 histories with 3 variants
Longitudinal stress histories	100 controlled 20-observation sequences	200 sequences
Initial expert response audit	100 common contexts across 3 systems	150 contexts across 3 systems

These are target counts, not existing resources. The smaller plan should be the default until annotation throughput is measured. Do not manufacture superficial variants to meet a count. If source/template grouping changes the exact split sizes, preserve disjointness and report the actual counts. The independent unit is the family, not each paraphrase or model run.

For each family, prepare four histories: correct work including alternative valid strategies; a persistent conceptual-error pattern; a transient-error pattern such as arithmetic, transcription, or language confusion; and an ambiguous pattern where multiple explanations remain plausible. Allocate transient categories across topics and deliberately include

assisted responses and contradictions. Histories may contain more than one observation when persistent-state evaluation requires it.

With four RAs, a 300-family target is substantial. Measure the median time for authoring, solving, trace design, and independent review on the first 20 families. Multiply the measured time by the target volume and add adjudication effort before promising the expanded plan. Record human curation hours as a real system-development cost.

7 3 Problem and trace fields

Record	Required information
Problem	Item ID, family ID, source/license, topic, difficulty, language, exact question, units, accepted answer forms
Solution	Reviewed strategies, step relations, assumptions, symbolic expressions, alternative paths, graph version
Attempt history	Ordered student events, original/normalized text, assistance events, timestamps, independent-evidence groups
Error annotation	First invalid step or no error, error category, interpretation uncertainty, valid alternative flag
Diagnosis annotation	Supported candidates, ruled-out candidates, unresolved flag, evidence sufficient to commit or not
Probe	Target ambiguity, allowed state, response categories, expected burden, disclosure review, branch records
Output policy	Permitted action, permitted facts, prohibited decisive disclosures, authorized-answer state

Keep latent simulator or author-intended labels in a separate evaluator-only file. The model-visible record must contain only the question, available history, and tools/reference information allowed for that condition. Strip diagnostic codes from filenames, prompt examples, answer strings, and metadata exposed to the tutor.

7 4 Annotation protocol

Two mathematically competent Bengali-speaking annotators should independently label every test history and every probe branch used for primary evaluation. A third reviewer resolves disagreements. Training data may use one annotator with sampled independent review if resources require it, but report that difference.

Calibrate the rubric on 30-50 development histories before final annotation. Include wrong percentage bases, correct algebra with an arithmetic slip, equivalent valid solutions, mixed numerals, an incomplete step, a copied answer, and ambiguous short responses. Record both initial disagreement and final adjudication.

Annotators must distinguish the earliest mathematical error from the cause of that error. When a student writes only a wrong option, a unique cognitive cause is often unknowable. Allow a candidate set and insufficient-evidence label. Do not force consensus on a specific misconception when the observable evidence cannot support it.

For controlled scenarios, the intended cause is part of the experimental construction and may provide a latent ground truth. Nevertheless, annotators should first see only the model-visible prefix and label what is supportable from it. Score both agreement with the constructed cause after diagnostic evidence and whether early commitment was justified. Do not reward lucky guesses on intentionally unidentifiable prefixes.

For authentic learner traces, use expert judgments of evidence support and later independent probes or transfer attempts. Do not describe the adjudicated label as direct access to the learner's true mental state. Report a clearly identified natural-data subset separately from authored or model-generated traces.

Report Cohen's kappa for two-annotator categorical judgments or Krippendorff's alpha when the design requires it, plus raw agreement and label prevalence. A planning target of at least 0.70 agreement coefficient can trigger rubric review, but is not a universal validity certificate. First-error annotations also need exact agreement and disagreement distance. Preserve ambiguity rather than deleting every difficult case to inflate agreement.

7 5 Prevent contamination

Assign related templates, source duplicates, translations, numerical variants, and paraphrases to one family before splitting. Use exact hashes, normalized expression structure, and similarity review to flag possible overlap. Human-review flagged pairs. A changed name or number does not make a training problem independent of its test template.

Create new test situations where possible and state that pretraining contamination cannot be ruled out for public exam questions. Keep test diagnoses and rubric decisions out of prompt development, calibration, probe fitting, and model selection. A fixed solution key supplied as a deployment resource is different from using test labels to train the method; document the distinction carefully.

For an application with an authored question bank, the tutor may have the current question's verified reference solution at inference. Give the same reference access to every baseline in the primary comparison. Evaluate automatically constructed or unavailable reference solutions in a separate generalization experiment. Do not advertise the authored-bank result as unrestricted tutoring on unseen mathematics.

For longitudinal experiments, split authentic learners by identity and keep all their sessions together. In controlled simulation, split scenario templates and response policies as well as problem families where feasible. Randomly splitting individual turns is insufficient.

7 6 External datasets

Use a held-out subset from BanglaMATH or BenNumEval to test mathematical interpretation and solving; retain the original task definition and license. Use the Stepwise Verification dataset for an English first-error check and MathDial or MathTutorBench for a bounded tutoring evaluation. If translating or adding diagnoses, label the result as a new annotated derivative. [R5, R8-R10, R18]

DAGGER's DistractMath-Bn and GSM-Plus-BN provide relevant perturbation contexts. They do not replace the evidence-history benchmark. Check overlap with MGSM/MSVAMP/GSM-derived training resources before treating them as independent external tests. Choose one manageable external check for the workshop rather than several underpowered evaluations. [R12, R13]

8 Baselines that a reviewer can trust

8 1 Primary system matrix

All primary matched baselines receive the same student-visible history, verified problem resource, permitted output actions, and maximum budgets. Their state and decision mechanisms vary. The weak direct-generation baseline is a useful anchor, but it cannot be the only comparison.

ID	System	Implementation and purpose
B0	Reviewed rule/template tutor	Fixed first-error hints and clarification rules; no generative diagnosis; efficiency floor
B1	Strong prompted tutor	One structured prompt with reference solution, permitted actions, and recent history
B2	Verify then generate	Same step verifier as GonitSathi; feedback grounded in its output; no persistent diagnostic controller
B3	Bayesian diagnostic tutor	Small likelihood model, calibrated beliefs, templates, and the same probe bank; essential non-LLM control
B4	Validated learner-state tutor	Central state, schema validation, contextual BKT-inspired updates, graduated hints; IntelliCode-inspired adaptation
B5	Plan and assessment memory	Step plan, progress, learner assessment, and action loop; ScaffoldLM-inspired adaptation
B6	Diagnostic reasoning workspace	Structured evidence and internal counterfactual validation; SLOW-inspired adaptation
G	GonitSathi controller	Governed evidence, revisable diagnostic state, bounded probes, same verifier and output policy

For B3, give the model the same observation categories and calibrated parameter budget as G wherever possible. If it matches G, report that result. To isolate whether an LLM is needed at all, use reviewed evidence features in one B3 condition and predicted features in another.

B4-B6 are explicitly named adaptations when the original models or tasks cannot be reproduced directly. Include a differences table covering training, task, model, tool access, and prompt changes. Do not put an adapted implementation under the original paper's name without qualification or imply that it reproduces its published score.

Start by reproducing the released implementation on its native task where feasible, then make the minimum documented domain changes. If native reproduction is not feasible, provide method-level fidelity checks and quote no reproduced performance. Publish our implementation and describe exactly which mechanism was retained.

8 2 Additional comparators

A no-guard direct-generation condition is useful for measuring the complete system's change in output risk, but keep the common guard fixed in mechanism comparisons. A random-question tutor is a weak educational baseline and must not replace B3-B6.

Use the same-family 8B model as an open-model reference for scale. Add GanitLLM or a released pedagogy-specialized checkpoint only when weights, license, and a runnable configuration are available. Treat a proprietary frontier model as optional context, using the exact model version and all call costs. The main claim should remain reproducible without a paid API. [R7, R11]

8 3 Prevent baseline handicaps

Allocate the same development optimization budget to the major methods, for example five documented prompt/policy revisions each. Use one common held-out development criterion. Do not tune G on teacher feedback while leaving a close baseline with its first generic prompt.

Use the same allowable reference content, input context cap, answer-release rules, and shared symbolic tools for the primary comparison. If a method naturally makes more calls,

show both its native configuration and a budget-matched adaptation. Removing essential steps to meet a tight cap changes the method; label that adaptation clearly.

9 Detailed run configuration

9 1 Model and inference setup

Use Qwen/Qwen3-4B as an initial primary open-weight backbone, with Qwen/Qwen3-1.7B and Qwen/Qwen3-8B for the scale comparison. These are a controlled same-family choice, not a claim that they are the newest or best models available. Verify the exact revision, license, tokenizer, and runtime before the experiment. Their official model cards document the releases and thinking-mode controls. [M1-M3]

Setting	Initial value or required record
Training	No backbone fine-tuning for the primary controller comparison
Precision	BF16/FP16 on workstation if it fits; separately report an explicit 4-bit deployment format
Context cap	4,096 input tokens per call, using the same truncation policy
Structured evidence output	Up to 256 generated tokens
Tutor response output	Up to 256 generated tokens
Thinking mode	Disabled consistently for the primary comparison; enabled only as a separate condition
Primary decoding	Greedy/deterministic where supported; record all settings
Stochastic robustness	Temperature 0.3, top-p 0.9, seeds 17, 29, 43 on a frozen subset
Episode length	Up to 6 tutor-student exchanges; final status includes unresolved/timeout
Diagnostic probes	Default maximum 1 per problem
Routine neural budget	Up to 2 calls per tutor turn; no silent retries
Tool budget	At most 4 symbolic checks per turn, with a recorded timeout
Failure handling	Explicit error record and reviewed fallback; all costs retained

These settings need a development smoke test before freezing. Compare methods under an identical total generated-token ceiling per episode, such as 3,072 tokens, as well as an equal-call condition. Calls and tokens are not interchangeable. Count any thinking tokens, repair calls, judge calls used by the deployed system, and larger-model escalations.

The observation normalizer and template controller can run on CPU. Use one available workstation for the main benchmark and report its exact CPU, GPU, RAM, operating system, and inference engine. A 24 GB GPU may be a practical candidate for these sizes, but measure actual peak memory rather than assuming every configuration fits.

9 2 Fixed turn evaluation

For the 300 core test histories, reset every method to the same allowed initial state and replay only the observed history. Ask it to produce its diagnostic belief, commitment decision, first-error decision, and next action. Do not let one method collect an additional student response in this test.

Run the eight primary systems once under deterministic decoding: 8 by 300 gives 2,400 evaluations. For stochastic robustness, use the same 100 frozen contexts and three seeds for a predeclared subset of methods. Repeated seeds quantify output variability; they do not create additional independent students or problem families.

Store both the proposed response and the actually released response. This distinguishes improvements in planning or generation from gains due to the shared release guard. Score the hidden diagnostic labels only after all outputs have been stored.

9 3 Interactive evaluation

Run the frozen 120 interactive scenarios with the same maximum six exchanges and one-probe allowance. Define the response bank before evaluating the tutors. Each bank branch must correspond to an eligible question or a recognized action category; unsupported actions receive a documented ambiguous/clarification response rather than a favorable invented continuation.

For eight methods and three stochastic scenario seeds, this gives 2,880 episodes. At six turns and two neural calls per turn, the upper bound is 34,560 tutor calls. Simulator calls, if used, are additional. Profile actual average lengths on development cases and estimate the workload before scheduling the full run.

Use common scenario seeds across methods and preserve the same latent student condition. Free-form LLM simulation should be a secondary check with at least two simulator families or styles and a simple scripted reference. Never use one model simultaneously as tutor, sole simulator, sole evaluator, and source of the diagnosis labels.

Because a probe itself can teach, compare equal-interaction and equal-time budgets. Keep pre-response and post-response diagnostics separate. A diagnostic gain from an additional response must not be described as a compute-only gain.

9 4 Run registry and checkpoints

Every run must have a configuration hash, code commit, dataset hash, prompt version, model revision, seed, hardware ID, start/end time, and status. Save raw predictions before scoring. Keep training, development, test, and simulated data in separate namespaces.

Do not rerun only failed or unfavorable examples and overwrite the first output. Register reruns, preserve the reason, and use one consistent retry rule for every system. A genuine software bug may require invalidating an entire affected run; document what changed and regenerate all affected comparisons.

10 Metrics and predeclared decision rules

10 1 Diagnostic risk and coverage

Define unsupported-commit risk as the number of active diagnostic commitments not justified by the adjudicated evidence divided by all active commitments. Define commitment coverage as eligible diagnosis opportunities receiving an active commitment divided by all eligible opportunities. Report counts and confidence intervals; risk is undefined when no claims are activated. Score episode-specific and persistent claims separately using the same eligible opportunities across systems.

Also report the unconditional unsupported-commit rate over all eligible opportunities. It is useful operationally but can be reduced simply by abstaining more. The primary comparison must therefore include a risk-coverage curve and at least one common attainable coverage level.

Choose a primary threshold on development data. A possible planning rule is: maximize coverage while keeping the upper 95% confidence bound on unsupported-commit risk below 5%. If no configuration satisfies the rule or the sample is too small to support it, report that fact and the full curve. The 5% value is an operating target, not a certified guarantee. Do not keep lowering coverage until the system appears reliable.

For primary pairwise inference, compare G with the strongest development-selected close baseline at a common coverage such as 60%, if both can attain it. Freeze that coverage before testing. Report the whole curve as sensitivity, and predeclare a minimum acceptable loss in useful action coverage.

10 2 Calibration and error localization

Metric	Operational definition and reporting rule
Multiclass Brier score	For each identifiable case, sum squared differences between candidate probabilities and the one-hot reference; average over cases; lower is better
Negative log likelihood	Negative log probability of the reference class; clip with a declared epsilon only for numerical computation
ECE	Confidence-versus-accuracy gap in predeclared bins; use 10 equal-frequency bins when sample size permits; show a reliability plot
Diagnostic macro-F1	Compute F1 per class and average equally on identifiable cases; declare abstain/unknown handling
Persistent-claim precision	Supported persistent misconception claims divided by persistent claims activated
First-error accuracy	Exact first-invalid-step match, including a no-error category
Valid-path acceptance	Accepted valid alternative solutions divided by valid alternatives
Unsupported-input rate	Unverifiable or unsupported inputs divided by all inputs; do not quietly omit them

Do not calculate a single-label Brier score against a forced label on a genuinely ambiguous human case. Report set-valued compatibility and justified abstention for those cases. For controlled latent-cause scenarios, label the Brier score as prediction of the constructed cause and separately score whether early evidence was sufficient to commit.

Mastery probability and misconception probability have different meanings. If BKT estimates future independent success, evaluate it on later independent success. Do not compare that number directly with G's probability of a specific misconception as though they were the same target.

10 3 Probe and intervention metrics

Measure diagnostic improvement after the probe using Brier change or reduction in adjudicated error, not only lower entropy. An overconfident wrong belief has low entropy. Report the proportion of probes that change the selected intervention, the proportion judged necessary, and response burden in seconds and extra turns.

Inappropriate-intervention rate is the proportion of contexts receiving an action that is mismatched to the supported diagnosis or violates the rubric. Score actions against an allowed set when several teaching moves are reasonable. A clarification can be appropriate without a firm diagnosis.

Measure local repair on an independent retry, and near-transfer success on a distinct problem. Keep unassisted success separate from assisted completion. Completion within a simulated conversation is not evidence of actual student learning.

10 4 Released response quality

Use an expert rubric with separate dimensions: mathematical correctness, targeting of the observed difficulty, consistency with the evidence, appropriate disclosure, clarity in Bengali, and actionable usefulness. Score each dimension on a documented 0-2 scale, with anchored examples. Report per-dimension results rather than only a summed score.

Define an accepted intervention as a released response that is mathematically acceptable, permitted by the disclosure policy, and useful for a plausible next student step. This joint outcome prevents empty but harmless fallback messages from looking successful. Report abstentions and unresolved cases separately.

Answer-leakage rate uses protected tutor turns as its denominator. Authorized worked solutions belong in a separate condition. Have experts inspect cumulative disclosure over a dialogue, not only individual responses. Measure false blocking and fallback usefulness as well as leakage. [R4]

10 5 Resource metrics

Report wall-clock latency per turn and episode, p50 and p95, time to first visible response if applicable, input/output/thinking tokens, neural calls, symbolic calls, cache hits, fallback use, model bytes, peak RAM/VRAM, and network bytes. State whether latency includes normalization, scheduling, verification, generation, and release auditing.

Report total compute or monetary cost per accepted intervention as total incurred cost divided by the number of accepted interventions; declare it undefined when none are accepted. Include rejected outputs and fallbacks in the numerator. Keep student waiting time separate from machine processing time.

For cloud pricing, record the provider price and date at run time. For local inference, do not label it free: report device time and, where available, measured energy. Do not infer energy savings or carbon reductions directly from token counts.

11 Main experiments

11 1 E1 Diagnostic reliability at matched resources

Run the fixed-history test on B0-B6 and G using the common 4B backbone where applicable. Give all methods the same verified reference solution, history, and inference budgets. Produce a main table with unsupported-commit risk, coverage, Brier, first-error accuracy, accepted intervention rate, calls, and p95 latency.

Select the strongest close comparator on development data, not after seeing test results. Report paired differences with family-clustered confidence intervals. The claim is supported only if improved diagnostic risk is not explained by collapsed coverage or usefulness. Include direct-generation B1 as context, but lead the discussion with B3-B6.

11 2 E2 Value of obtaining one diagnostic response

Compare no probe, a generic clarification, a random eligible probe, a fixed expert-selected probe, and the information-gain-selected probe. Keep maximum additional responses and response time comparable. Use the same hypothesis prior, response bank, and release policy.

Measure outcomes before and after the response. Analyze identifiable and persistently ambiguous cases separately. Add a placebo condition that spends an extra LLM call on self-reflection without receiving new student evidence. This separates the value of an observation from the value of more model computation.

If a fixed expert probe matches information-gain selection, simplify the paper's method claim. If probing reduces uncertainty but not diagnostic error or useful action quality, do not claim the diagnostic policy works.

11 3 E3 Model scale and compute efficiency

Run B1, B3, the strongest memory/workspace baseline, and G at 1.7B, 4B, and 8B where each model-based component is applicable. Keep prompts and protocol fixed except for required tokenizer/runtime adaptation. Compare 4-bit and higher precision on a held-out subset without mixing precision changes into the primary algorithm comparison.

Plot accepted intervention quality and diagnostic risk against measured latency and token cost. Label every point with model, precision, and budget. A smaller G system matching a larger baseline is an empirical finding only if both were evaluated on the same examples and reference-access regime with adequate uncertainty intervals.

Do not claim non-inferiority from a nonsignificant difference. If using a non-inferiority claim, specify a justified margin in advance and show the confidence interval for the quality difference relative to that margin. Otherwise describe the observed trade-off without an equivalence claim.

11 4 E4 Assistance and conflicting history

Construct paired histories with identical final answers but different provenance: unassisted work versus key-step assistance, copied solution versus independent transfer, and repeated attempts from one problem versus independent problems. Measure changes in commitment, later prediction, and false mastery.

For contradiction recovery, use controlled sequences: several independent observations support a claim, one transient slip occurs, a contradictory valid method appears, or a series of new independent successes follows instruction. Track unsupported commitments and observations needed to contest or retract a false claim.

Run 20-observation stress histories to reveal accumulation errors. Report state transitions and error persistence across the sequence, with sequence-level confidence intervals. Simulated history length is not elapsed real-world retention time.

11 5 E5 Language and distribution stress

Create three meaning-preserving variants of 100 test histories: Bengali/Arabic numeral changes; code mixing or an expert-reviewed paraphrase; and irrelevant numerical context. Keep the mathematical answer and intended evidence the same. Have a reviewer verify that the perturbation did not change the task.

Report paired performance change by perturbation type and topic. Cluster variants under their original family. Include valid alternative solutions and missing/ambiguous steps. Inspect whether failures arise in normalization, semantic interpretation, diagnosis, or realization rather than attributing every drop to the backbone.

Add a leave-one-topic-out or stricter new-template check only if the development budget permits it. Keep it separate from the primary within-scope benchmark and do not tune after observing its outcomes.

11 6 E6 Resource limited operation

Run the local controller with the network disabled, the same configuration online, and an explicitly labeled hybrid route if implemented. Also test a constrained connection and an intermittent outage with disclosed settings. Verify that offline runs cannot silently call an API.

Measure completed supported actions, response quality, queued sync events, duplicate prevention, and time to recovery. If only the question packs and templates run on-device while the model runs on a workstation, describe it as offline content plus remote inference, not fully offline neural tutoring.

For a mobile claim, benchmark an actual named device with recorded RAM, chipset, OS, model format, runtime, and thermal conditions. Repeat at least 30 measured interactions per relevant mode after warm-up. Measure cold start separately. Battery measurements require a reproducible power method and idle baseline; otherwise leave energy unclaimed.

12 Ablations that isolate the mechanism

12 1 Core component ablations

Change one mechanism at a time, keep other resources fixed, and state the prediction before running the experiment. Reuse saved evidence features when testing only the controller so that extraction randomness does not confound the result. Then confirm the most important differences in the end-to-end pipeline.

ID	Ablation	Held constant	Mechanism tested and outcomes
A1	Direct candidate-to-state update	Same extracted candidates and history	Evidence admission: unsupported commits, risk/coverage
A2	Schema validation only	Single writer and format checks retained	Semantic evidence checks beyond software validity
A3	Ignore assistance provenance	Same student text and correctness	False mastery and later independent success prediction
A4	Count every event independently	Same observations	Duplicate/correlated-evidence inflation
A5	Overwrite conflicting evidence	Same current observation	False-claim persistence and correction recovery
A6	Use original heuristic product	Same factors and observations	Need for calibrated likelihood or classifier model
A7	Disable diagnostic probes	Same neural budget	Contribution of new observed evidence
A8	Random eligible probe	Same probe count and bank	Value of information-based selection
A9	Generic clarification	Same additional interaction budget	Value beyond merely asking one more question
A10	Always generate a hint	Same state and action policy	Resource router's cost and usefulness trade-off

ID	Ablation	Held constant	Mechanism tested and outcomes
A11	Remove output guard	Same precomputed candidate texts	Contribution and cost of release verification
A12	Replace realizer with reviewed templates	Same actions and state	Whether generative realization is necessary

Run A1-A9 on the fixed or interactive layer appropriate to the mechanism. A11 is an offline audit condition; its unsafe outputs need not be shown to learners. Removing a guard should not change the underlying proposed text in a release-attribution comparison.

The literal zero-on-wrong-work reliability factor from the report may be included as a clearly labeled flawed-rule diagnostic, but it must not be the principal baseline that G defeats. Compare against the corrected heuristic and strong probabilistic baseline as well.

12 2 Factorial test for the central interaction

Use a two-by-two design crossing governance on/off with bounded probing on/off. The four conditions are direct state without probes, governed state without probes, direct state with probes, and governed state with probes. Share the extraction model, current belief inputs, probe candidates, and resource cap.

Compare the change produced by adding probes under governance with the change produced by adding probes without governance. Estimate the interaction using paired family-clustered resampling or an appropriate repeated-measures model. A full-system gain alone does not show that the two mechanisms reinforce one another.

When a policy makes different decisions because its state differs, keep that as part of the end-to-end intervention. Also run a replay condition with frozen candidate evidence and actions to isolate the update rule. Explain the difference between these two estimands.

12 3 Confound controls and oracles

Potential explanation for a gain	Required control
More model computation	Equal-call and equal-token comparisons; extra-call self-reflection placebo
More student time	Generic/random probe and equal-time conditions
More abstention	Common-coverage risk and useful-action coverage
Privileged solutions	Same verified references for all primary baselines
Better evidence extractor	Frozen shared-extraction comparison plus end-to-end comparison
Output rejection instead of better planning	Score proposed and released responses; common-guard condition
Easier test items routed to cheap model	Report quality and item mix by route; compare identical held-out inputs
Convenient synthetic student behavior	Scripted bank, second simulator family, and natural/expert-reviewed subset

Use oracle first-error labels to estimate losses due to the verifier, oracle response likelihoods in constructed scenarios to estimate probe-model losses, and oracle diagnoses to bound

downstream action/realization quality. Label every oracle as an upper-bound diagnostic, never a deployable competitor. The main results must use predicted inputs.

12 4 Sensitivity and efficient scheduling

After the primary ablations, vary activation thresholds, entropy threshold, maximum probe count, allowed history length, candidate probe count, response-model smoothing, and precision. Use development data to select operating points and a small predeclared test grid for robustness reporting.

Perturb response likelihoods or flatten them toward a uniform distribution to test whether information-gain selection is fragile. Remove recency weighting separately from evidence provenance. Distinguish source credibility from observation correctness, and do not assume longer response time proves weaker knowledge.

Avoid a full Cartesian product over every setting. First run a 20-case development smoke test, then a frozen 100-case mechanism screen, then the essential primary comparisons on the complete held-out set. The screening set must be development data if it affects what the team tunes or promotes to a headline claim.

13 Human evaluation and learning studies

13 1 Expert audit for the workshop paper

Use three systems on 100 common contexts: G, the strongest close baseline selected on development data, and B3 or the best reviewed template condition. This produces 300 released responses. Two independent raters produce 600 response-level judgments; use a third rater for disagreements. Expand the sample only after checking available reviewer hours.

Randomize system order, hide names and model identities, and remove identifying formatting. Give raters the problem, observed student history, relevant reference solution, and permitted action policy. For the evidence-support dimension, hide the system's own explanation of why its diagnosis is correct until the initial judgment is complete.

Evaluate some complete dialogues as well as fixed turns so cumulative answer disclosure is visible. Predefine the context sampling by topic, ambiguity, assistance, and system failure status. A representative sample estimates overall quality; an oversampled failure audit should be reported separately or appropriately weighted.

Calibrate the rubric on 20 development examples. Ask raters to identify the evidence span supporting a diagnosis, the mathematical issue in an incorrect reply, or the content causing leakage. A score without a traceable rationale is difficult to audit. Report agreement before adjudication and both per-dimension and joint accepted-intervention results.

An LLM judge can screen large volumes or provide a secondary score. Validate it against held-out human labels and report false positives, false negatives, and agreement by language and error type. Do not use the same model's opinion as the sole evidence that its own state governance works.

13 2 Optional short learner pilot

If recruitment and the applicable institutional ethics process are ready, use a 24-40 adult-learner feasibility pilot. Assess whether the probes are understandable, how long learners wait, whether they feel repetitive, and whether the interface records assistance correctly. A brief independent pre/post or transfer check can be exploratory.

Use consent, pseudonyms, a clear withdrawal process, and a data-retention plan. Recruitment through the RA team's own classes must not affect grades or create pressure to

participate. Avoid collecting unnecessary personal or exam-identifying information. Authenticate and restrict access to any real learner-state store.

A small pilot generally cannot establish durable learning improvements. Do not call it an RCT showing efficacy merely because learners were randomized. If comparing interfaces within learners, counterbalance topic/order, use different matched problems, and analyze learner-level dependence. If instruction from one condition can carry into another, a parallel-arm study is cleaner for learning outcomes.

13 3 Later efficacy study

For a later learning paper, compare two arms: G and the strongest practical tutoring baseline. Match problem coverage, session duration, available content, and access to worked solutions. Randomize learners, preferably stratifying by baseline proficiency, and keep assessment scorers blind. A four-arm study requires a larger sample and is not necessary to establish the controller's first effect.

Use independent pretest, immediate posttest, and delayed assessments on matched but unseen problems. Predeclare the primary outcome, for example posttest performance adjusted for pretest. Record adherence, time on task, assistance, and attrition. Near-transfer and delayed retention should be separate outcomes.

Plan sample size from a justified smallest meaningful effect. As a rough two-arm calculation, detecting a standardized mean difference of 0.4 with 80% power and a two-sided 5% test requires about 100 completers per arm; allowing 20% attrition suggests about 250 recruited in total. This is an illustrative planning calculation, not a final power analysis. Baseline adjustment, clustering, attrition, and multiple outcomes can change the requirement.

Analyze by randomized assignment where appropriate and describe missing-data assumptions. Prefer adjusted posttest differences and confidence intervals over relying only on normalized learning gain, which can be unstable for high pretest scores. Four- and eight-week retention assessments must occur four and eight weeks after the intervention, not after recruitment.

14 Statistics and interpretation

14 1 Define the estimand before the run

The primary estimand is the difference in unsupported-commit risk between G and the strongest prespecified close baseline at comparable diagnostic coverage on held-out problem families. State whether each metric averages over families, episodes, or turns. Use macro-averaging when a few long histories would otherwise dominate.

For fixed histories, use paired bootstrap confidence intervals with 5,000 resamples of whole problem families, retaining all variants and histories within each sampled family. For simulated longitudinal histories, resample whole sequences or their independent generating family. With real learners and repeated items, use a crossed-effects or suitable multiway-clustered analysis that respects both learner and item dependence.

Treat multiple generation seeds as repeated measurements. Average within the independent unit or use a hierarchical model; do not multiply the sample size by the number of seeds. Report between-seed variability separately from uncertainty about new families or learners.

14 2 Tests and reporting

Report effect sizes and 95% confidence intervals for the primary paired differences. A paired permutation test can supplement them when its exchangeability assumptions hold.

McNemar's test is suitable for independent paired binary units; it must not be applied naively to hundreds of correlated turns and paraphrases.

Keep one primary outcome/comparator. Use a predeclared Holm correction for a small confirmatory family of secondary tests, or label the ablation analyses exploratory and emphasize intervals. Do not search among metrics, model sizes, thresholds, and seeds for the smallest p-value.

For latency, report distributions and paired differences on a controlled workload. Run warm-ups, randomize/interleave method order where possible, and record caching and thermal conditions. Keep cold-start, warm inference, and network effects separate.

If no failures are observed, report the tested denominator and an upper confidence bound. Zero observed failures is not proof of zero risk. For independent trials, the approximate rule of three gives a 95% upper failure rate near three divided by the number of trials; correlated histories require a more appropriate interval and should not use that shortcut as a guarantee.

14 3 Decision rules for claims in the paper

Observed result	Defensible interpretation
Lower risk with similar coverage and useful actions	Supports improved selective diagnostic reliability in the evaluated setting
Lower risk with much lower coverage	Shows a conservative policy trade-off; does not establish superior diagnosis
Selected and random probes perform similarly	Extra interaction may help; information-gain selection is not supported
B3 matches or beats G	Simple probabilistic control may be sufficient; simplify the contribution
Small G matches larger baseline with a narrow interval	Supports the measured quality-cost comparison; stronger equivalence claims need a predefined margin
Output guard reduces leakage but raises empty fallbacks	Report the safety-usefulness trade-off and failures
Simulator gains with no human evidence	Supports simulator performance, not learner efficacy
New labels fail on held-out topics	Limit the scope and report the transfer failure

Include representative failure cases selected by a predefined taxonomy rather than only dramatic anecdotes. Cover semantic parsing failure, correct alternate strategy rejected, wrong diagnostic commitment, misleading probe, overconfident update, stale evidence, excessive fallback, cumulative leakage, and unsupported input.

For each failure, show the relevant student evidence, system decision, expected decision, cause, and whether the error persisted. Keep complete traces in the appendix or artifact with personal data removed.

15 RA assignments and schedule

15 1 Ownership with independent review

Role	Primary responsibilities	Required independent check
RA 1 Research and paper lead	Literature matrix, claim ledger, manuscript, reference verification	RA 4 checks cited claims and submission rules
RA 2 Data and annotation lead	Problem families, trace/probe bank, splits, rubrics, adjudication logistics	RA 1 checks disjointness and annotation assumptions
RA 3 Methods lead	Normalizer, verifier, state controller, probe policy, shared interfaces	RA 2 checks outputs against reviewed examples
RA 4 Evaluation lead	Baselines, experiment registry, metrics, uncertainty, profiling, reproduction	RA 3 checks scoring and budget accounting
Supervisor	Scope, novelty, key protocol choices, final claims and submission readiness	Team supplies concrete evidence at each gate

Roles can be combined for a smaller team, but the author of a component should not be its only evaluator. Keep one coordinator for the frozen dataset and one for the main experiment registry.

Each progress update should include a completed artifact, one measured result or resolved decision, an unresolved issue, and the next deliverable. Avoid updates consisting only of papers read, agents added, or code written. Bring a concrete failure trace when asking for methodological help.

15 2 Calendar to the submission deadline

Dates	Deliverable	Completion gate
12-15 September	Close-paper review, claim audit, 20-family annotation pilot, method specification	Contribution and annotation feasibility approved
16-22 September	Core dataset authoring; verifier/controller prototype; B0-B3	End-to-end development example with full logs
23-28 September	Independent test annotation; B4-B6 adaptations; pilot profiling	Split audit, rubric agreement, baseline fidelity recorded
29 September-5 October	Freeze protocol; primary runs; probe and core ablations	Raw outputs and complete result tables available
6-10 October	Expert audit, scale/robustness subset, uncertainty analysis	Main conclusion survives strong comparisons
11-14 October	Full manuscript, figures, limitations, reproducibility package	Independent RA reproduces one principal table

Dates	Deliverable	Completion gate
15 October	Anonymous review and page-limit audit	Claims match evidence; files open and compile
16 October	Internal submission target	Supervisor checks final concrete submission package
17-18 October	Deadline buffer and necessary corrections	Portal receipt and uploaded version verified

Do not start the extended RCT merely to fill the experiment section. If annotation or baseline reproduction is delayed, reduce topic coverage transparently, switch to the short-paper scope, or omit unsupported secondary claims. Preserve the strongest comparator and primary causal ablations before adding more model names.

15 3 Supervisor checkpoint package

Before large runs, submit a two-page protocol containing the contribution, strongest competing explanation, exact primary metric, baseline list, independent sample unit, split hashes, operating-point selection rule, and compute estimate. Attach 10 representative annotated examples and two complete end-to-end traces.

At the results checkpoint, submit the full table, confidence intervals, failed runs, cost totals, all primary ablations, and at least five failure cases. At the manuscript checkpoint, every claimed improvement should point to a specific table or figure and an experiment ID.

16 Write the eight page paper

16 1 Title and abstract

Candidate title: GonitSathi Evidence Governed Diagnosis for Efficient Bengali Tutoring Agents.

A more targeted title after positive probe results is GonitSathi Bounded Diagnostic Probing for Reliable Small Model Tutoring. Choose the title after the strongest result is known; do not put unmeasured learning gains, universal reliability, or edge deployment in the title.

Build the abstract in five moves: the specific diagnostic problem; why current memory or tutoring controls leave the evaluated issue unresolved; the implemented method; the dataset and comparison design; and the measured result with its main limitation. Keep every number tied to a completed run.

Abstract drafting scaffold: Small open-weight tutors can interpret a student's isolated error as evidence of a persistent misconception. We study this problem in Bengali competitive mathematics using an evidence-governed controller that retains observation provenance, delays unsupported commitments, and selectively obtains one diagnostic response. We evaluate the controller on our reviewed benchmark against matched probabilistic and state-memory baselines. Insert the measured risk difference, coverage, intervention quality, and cost here. Conclude with the evaluated scope and the strongest remaining limitation.

This scaffold is deliberately not a finished abstract. Replace the result sentence only after experiments are complete, and revise the problem framing if a close baseline solves it equally well.

16 2 Page allocation

Component	Approximate pages	Purpose
Abstract	0.25	Problem, method, evidence, result
Introduction	1.00	Concrete failure and narrow contributions
Related work	0.75	Closest architectural and Bengali/efficiency comparisons
Method	1.75	Evidence model, state policy, bounded probing, release path
Experimental setup	1.25	Dataset, splits, baselines, resource budgets, metrics
Results and ablations	2.25	Primary evidence, causal tests, efficiency, failures
Limitations and ethics	0.50	Scope, diagnostic uncertainty, annotation and deployment limits
Conclusion	0.25	What the experiments establish
Total main text	8.00	References and appendices outside full-paper limit

For a four-page submission, retain the primary problem, method, one main comparison, the central two-by-two ablation, and limitations. Move implementation and rubrics to supplementary material only when the portal permits it. Do not remove the strong baseline and leave a visually impressive architecture with no decisive experiment.

16 3 Essential figures and tables

Figure 1 should show the event-to-state boundary and the branch between probing and teaching. Label what is observed, what is inferred, what is verified, and where the LLM lacks write authority. Avoid a decorative diagram with many unspecified agents.

Figure 2 should show diagnostic risk against coverage for G and the closest baselines. Figure 3 should show useful intervention quality against measured resource cost across model sizes or budgets. If space is tight, combine related plots into readable panels, keeping axis definitions and denominators clear.

Table 1 should describe the benchmark and independent units. Table 2 should report the principal matched comparison. Table 3 should report the central governance/probing ablation and assistance control. Put full class-level results, model configurations, and secondary sensitivities in the appendix.

16 4 Related work and claim writing

Organize the related-work section around state reliability, diagnostic interaction, and efficient Bengali tutoring. Compare mechanisms and evaluation protocols rather than writing one paragraph per paper. Cite the closest overlap next to the specific claim.

Use language such as our experiments test, the controller reduces the observed rate, and the result holds on the supported benchmark. Replace first, guaranteed, eliminates hallucination, human-like understanding, and improves learning with narrower wording unless the paper provides the required evidence.

Separate model capacity from framework contribution. If a specialized backbone is stronger, report its gain separately. If the framework needs extensive expert curation, report those hours and discuss the limits to new-topic scaling. Open weights alone do not make all code, data, licenses, and annotations open.

16 5 Limitations that belong in the paper

State that diagnostic labels are evidence-based judgments rather than direct mental-state observations. Explain the role of authored problems, reference solutions, constructed histories, and any LLM simulation. Discuss rare misconceptions, multiple simultaneous causes, alternative valid solutions, response-model misspecification, and the possibility that probes teach as well as diagnose.

Report the domain and language coverage, hardware actually tested, annotation cost, and unsupported inputs. State whether learner gains and long-term retention were measured. If not, present them as untested. These boundaries make the paper assessable and do not undermine a well-supported methodological contribution.

17 Reproducibility and final submission gates

17 1 Artifact contents

Release the data schema, allowable data, split manifest, annotation rubric, adjudication procedure, reviewed solution/probe resources, model and prompt configurations, controller code, baseline adaptations, raw anonymized outputs, scoring scripts, cost logs, and figure/table generation commands. Provide a small smoke-test example and a full-run command.

Keep dependencies pinned and record model revisions rather than only display names. Include inference settings, quantization method, device specifications, and execution dates. Provide licenses and source attribution. If original exam text cannot be redistributed, release permissible annotations, identifiers, or replacement examples and explain the access procedure.

A second RA should run a clean reproduction of one primary table from saved outputs and at least a small inference subset from scratch. Record expected runtime and the differences that hardware nondeterminism can cause. Reproduction success means the reported values can be regenerated, not merely that the repository imports.

17 2 Scientific gate

1. The contribution is stated in one sentence and acknowledges the closest 2026 papers.
2. Test examples, families, and authentic learners are separated correctly from development.
3. Gold diagnoses and hidden simulator state are inaccessible to the tutor.
4. G is compared with a strong Bayesian controller and at least one credible memory/workspace baseline.
5. Reference-solution access, budgets, and output guards are controlled and disclosed.
6. Diagnostic risk is reported with coverage and usefulness; unresolved cases are retained.
7. Probing is compared with equal-interaction and extra-compute controls.
8. No unperformed learning, retention, safety, or mobile-deployment result is claimed.

17 3 Manuscript and upload gate

1. Compile using the workshop-linked ACML style and verify the correct review settings.

2. Check the eight-page full-paper limit and the portal's current supplementary rules.
3. Remove identifying author blocks, affiliations, acknowledgments, PDF metadata, file paths, and revealing repository details where required for anonymity.
4. Handle self-citations neutrally and according to venue rules; do not remove relevant prior work merely because the team authored it.
5. Verify every bibliographic title, author list, year, venue, URL, and quantitative statement against its source.
6. Ensure tables distinguish measured values, simulation results, unavailable values, and proposed work.
7. Check that the abstract's numbers agree exactly with the final table and confidence intervals.
8. Open the final PDF and supplementary archive on another machine; verify embedded fonts and all figure labels.
9. Recheck workshop-specific policies and submit through the correct OE-Agent group after the supervisor reviews the complete package.
10. Verify the uploaded version and receipt; preserve the final PDF, source archive, configuration hashes, and receipt together.

Appendix A Event schema and interface contracts

A 1 Example observation record

The values below illustrate a development record; they are not a student dataset or an experimental result. In this example, the student's relation is invalid, but the captured and interpreted observation is usable diagnostic evidence.

```
{
  "event_id": "dev_event_0042",
  "learner_id": "participant_pseudonym",
  "item_id": "pct_012",
  "family_id": "inverse_budget_03",
  "independent_evidence_group": "attempt_012_a",
  "event_type": "student_step",
  "original_text": "new_quantity = 100 - 25",
  "normalized_text": "new_quantity = 100 - 25",
  "assistance_level": "none",
  "source_type": "independent_practice",
  "interpretation_status": "supported",
  "mathematical_status": "invalid",
  "verification_reason": "fixed_budget_relation_violated",
  "candidate_causes": ["percentage_base", "transient_slip"],
  "evidence_admitted": true,
  "claim_status": "provisional",
  "controller_version": "record_exact_commit"
}
```

In production, add a timestamp, consent/data-access metadata, tool version, model revision, and links to the relevant facts. Keep labels such as evaluator cause and adjudicated commitment support in an evaluator-only record.

A 2 Controller pseudocode

```

record student_event with immutable provenance
normalize while retaining original text
align and verify the interpreted mathematical relation
if interpretation is unresolved:
    log evidence without updating the diagnostic belief
    choose clarification or a supported fallback
else:
    form joint evidence features with assistance status
    reject duplicate evidence contributions
    update the calibrated diagnostic belief
    revise provisional, active, or contested claims
    if uncertainty, expected value, and budgets permit:
        select an eligible reviewed diagnostic probe
    else:
        select an evidence-compatible teaching action
realize action using a template or allowed model route
audit proposed text against facts and disclosure policy
release approved text or reviewed fallback
wait for actual student evidence before the next update

```

A 3 Minimal meaningful verification cases

Before experimental runs, verify that a reliably observed wrong step can update the error belief; an unresolved parse cannot activate a claim; an assisted correct answer is distinguished from independent mastery; duplicate events do not multiply evidence; a contradiction can contest and retract a claim; and the probe budget cannot be exceeded.

Also check alternative valid solutions, Bengali numeral preservation, timeouts, authorized answer release, cumulative leakage examples, and network-disabled operation. These are checks of the research mechanism and measurement integrity. Keep them small and tied to an identified failure risk.

Appendix B Prompt and evaluation contracts

B 1 Evidence extractor

Instruction to the model: read the supplied problem and observed student history. Identify the student's stated relation, relevant quantities, and evidence spans. Return candidate error explanations and uncertainty. Distinguish missing information from incorrect reasoning. Do not infer a permanent learner trait from one option. Do not create observations, edit persistent state, or use evaluator-only labels.

Required output fields: interpreted relation; evidence spans; interpretation uncertainty; candidate explanations; assistance references; suggested need for clarification. Mathematical status should come from the designated verifier rather than an unsupported assertion in the extractor output.

B 2 Response realizer

Instruction to the model: express the selected instructional action in clear Bengali using only the permitted facts. Respect the specified disclosure level. Ask at most the allowed number of questions. Do not change the diagnosis or action. Do not disclose the final answer or decisive step in a protected hint state. If the provided information is insufficient, return the designated fallback signal.

Record the exact final prompt, not only these prose instructions. Version Bengali translations and any few-shot examples. Use examples from training data and keep prompt content constant across matched comparisons except for the intended mechanism.

B 3 Expert rubric anchors

Dimension	Score 0	Score 1	Score 2
Mathematical correctness	Incorrect or misleading	Partially correct or insufficiently supported	Correct within the supplied assumptions
Evidence alignment	Unsupported diagnosis/action	Plausible but too strong or weak	Calibrated to the observed evidence
Targeting	Misses the relevant difficulty	Partly relevant	Addresses the earliest useful issue
Disclosure policy	Prohibited decisive disclosure	Borderline or over-restrictive	Appropriate for the authorized action
Bengali clarity	Confusing or materially mistranslated	Understandable with friction	Clear and natural for the intended learner
Usefulness	No viable next step	Some support but limited progress	Enables an appropriate next learner action

Define accepted-intervention thresholds before annotation. For example, require mathematical correctness and disclosure policy to score 2, evidence alignment at least 1, and usefulness at least 1. Report the strict all-dimensions version as a sensitivity check if justified, without choosing the version that makes G look strongest.

Appendix C Result tables to prepare before experiments

C 1 Main diagnostic and efficiency table

Create one row per system and model. Required columns: independent family count; unsupported-commit risk with 95% interval; commitment coverage; Brier on identifiable cases; first-error accuracy; accepted-intervention rate; mean neural calls; total generated tokens; p50/p95 latency; peak memory. State all denominators in the caption.

Use NR to mean not run and NA to mean not applicable. Never place zero in a missing-result cell. For a template tutor without a probabilistic diagnosis, Brier is NA; it can still be evaluated for action quality and efficiency.

C 2 Governance and probing table

Condition	Commit risk and coverage	Diagnostic change after response	Useful action quality	Calls and extra turns
No governance and no probe	NR	NA	NR	NR
Governance without probe	NR	NA	NR	NR
Probe without governance	NR	NR	NR	NR

Condition	Commit risk and coverage	Diagnostic change after response	Useful action quality	Calls and extra turns
Governance with selected probe	NR	NR	NR	NR
Governance with random probe	NR	NR	NR	NR
Governance with generic clarification	NR	NR	NR	NR

C 3 Per run manifest

Record experiment ID, system ID, backbone revision, controller commit, prompt hash, dataset/split hash, precision, runtime, device, seed, limits, start/end time, number of eligible units, completed units, failures, fallback count, output file, and scoring version. Attach the manifest to every result table so that a reviewer can trace each reported number.

Appendix D Reference guide

The primary sources below support the literature comparisons and venue information. Source-reported performance is not treated as a GonitSathi result. Reference descriptions are reading guidance; the team should retrieve official bibliography records for the final manuscript.

Workshop and project sources

[W1] OE-Agent at ACML 2026. Open and Efficient Agentic AI for Resource-Constrained Environments. Workshop scope, submission rules, and dates. Checked 12 September 2026. [Workshop website](#).

[W2] OE-Agent OpenReview group. Submission portal and event listing. [OpenReview](#). Main-conference information and the linked style files are available through the [ACML 2026 call for papers](#).

[S1] EduTech and GonitSathi Research Groups. Epistemically Governed Neuro-Symbolic Tutoring: Transitioning from Deterministic Semantic Compilers to GonitSathi-BCS. September 2026. Supplied project report, GonitSathi_Research_Report_v2.pdf, 17 pages. Read pp. 2-3 for prior-engine claims, pp. 10-12 for the proposed evidence and probe rules, and pp. 14-15 for feasibility and the proposed field study.

Closest tutoring systems and evaluations

[R1] David, J., and Ghosh, S. 2026. IntelliCode: A Multi-Agent LLM Tutoring System with Centralized Learner Modeling. EACL System Demonstrations, pp. 129-138. [Paper and bibliography](#).

[R2] Li, Z., Zhu, Q., Wang, M., Li, J., and Huang, H. 2026. Planning-Guided Tutoring with Assessment-Driven Memory for Pedagogical LLM Tutors. ACL, pp. 7165-7188. [Paper](#). [Author code repository](#).

[R3] Wei, Y., Li, R., and Jiang, B. 2026. SLOW: Strategic Logical-inference Open Workspace for Cognitive Adaptation in AI Tutoring. AIED 2026; arXiv:2603.28062. [Accessible paper](#). [AIED proceedings listing](#).

[R4] Kadir, N. 2026. Auditable Release Control for Pedagogical Leakage in LLM Tutors. arXiv:2608.00515, submitted 1 August 2026. Preprint, not peer reviewed. [Paper](#).

[R5] Daheim, N., Macina, J., Kapur, M., Gurevych, I., and Sachan, M. 2024. Stepwise Verification and Remediation of Student Reasoning Errors with Large Language Model Tutors. EMNLP, pp. 8386-8411. [Paper and dataset](#). [Author code](#).

[R6] Puech, R., Macina, J., Chatain, J., Sachan, M., and Kapur, M. 2025. Towards the Pedagogical Steering of Large Language Models for Tutoring: A Case Study with Modeling Productive Failure. Findings of ACL, pp. 26291-26311. [Paper](#).

[R7] Dinucu-Jianu, D., Macina, J., Daheim, N., Hakimi, I., Gurevych, I., and Sachan, M. 2025. From Problem-Solving to Teaching Problem-Solving: Aligning LLMs with Pedagogy using Reinforcement Learning. EMNLP, pp. 272-292. [Paper](#).

[R8] Macina, J., Daheim, N., Hakimi, I., Kapur, M., Gurevych, I., and Sachan, M. 2025. MathTutorBench: A Benchmark for Measuring Open-ended Pedagogical Capabilities of LLM Tutors. EMNLP, pp. 204-221. [Paper](#).

Bengali reasoning and efficient agents

- [R9] Prama, T. T., Danforth, C. M., and Dodds, P. 2025. BanglaMATH: A Bangla benchmark dataset for testing LLM mathematical reasoning at grades 6, 7, and 8. MathNLP, pp. 134-149. [Paper](#).
- [R10] Ahmed, K., Osama, M., Sharif, O., Hossain, E., and Hoque, M. M. 2025. BenNumEval: A Benchmark to Assess LLMs' Numerical Reasoning Capabilities in Bengali. Findings of ACL, pp. 17782-17799. [Paper](#).
- [R11] Dipta, S. R., Mahbub, K., and Najjar, N. 2026. GanitLLM: Difficulty-Aware Bengali Mathematical Reasoning through Curriculum-GRPO. arXiv:2601.06767v3, revised 19 April 2026; the author record reports acceptance at ACL 2026 Findings. [Paper](#). Use the final proceedings citation if available when submitting.
- [R12] Al Nazi, Z., Dipta, S. R., and Kar, S. 2026. DAGGER: Distractor-Aware Graph Generation for Executable Reasoning in Math Problems. arXiv:2601.06853. [Paper](#). [Author project page](#).
- [R13] Paul, B., Mayouree, N. J., Karim, M. A., Nath, S. C., and Kundu, S. 2026. GSM-Plus-BN: A Perturbation-Based Benchmark for Bangla Mathematical Reasoning in Large Language Models. arXiv:2607.13248, July 2026 preprint. [Paper](#).
- [R14] Belcak, P., Heinrich, G., Diao, S., Fu, Y., Dong, X., Muralidharan, S., Lin, Y. C., and Molchanov, P. 2025. Small Language Models are the Future of Agentic AI. Position paper, arXiv:2506.02153v2. [Paper](#).
- [R15] Ong, I., and colleagues. 2025. RouteLLM: Learning to Route LLMs from Preference Data. ICLR 2025; arXiv:2406.18665 uses the title wording with Preference Data. [Conference record](#). [Author implementation](#).

Foundations and model records

- [R16] Gao, L., and colleagues. 2023. PAL: Program-aided Language Models. ICML, PMLR 202, pp. 10764-10799. [Paper](#).
- [R17] Corbett, A. T., and Anderson, J. R. Knowledge tracing: Modeling the acquisition of procedural knowledge. User Modeling and User-Adapted Interaction, volume 4, pp. 253-278. The publisher record lists December 1994; some later citations use 1995. Use the publisher's bibliographic record consistently. [Publisher page](#).
- [R18] Macina, J., Daheim, N., Chowdhury, S. P., Sinha, T., Kapur, M., Gurevych, I., and Sachan, M. 2023. MathDial: A Dialogue Tutoring Dataset with Rich Pedagogical Properties Grounded in Math Reasoning Problems. Findings of EMNLP, pp. 5602-5621. [Paper](#).
- [M1] Qwen team. Qwen3-1.7B official model record. [Model card](#).
- [M2] Qwen team. Qwen3-4B official model record. [Model card](#).
- [M3] Qwen team. Qwen3-8B official model record. [Model card](#).

Before submitting, export complete BibTeX from the official records, resolve version differences, and check all cited code/data links. The research claim should remain valuable if a stronger backbone replaces today's model choice.